HTTP

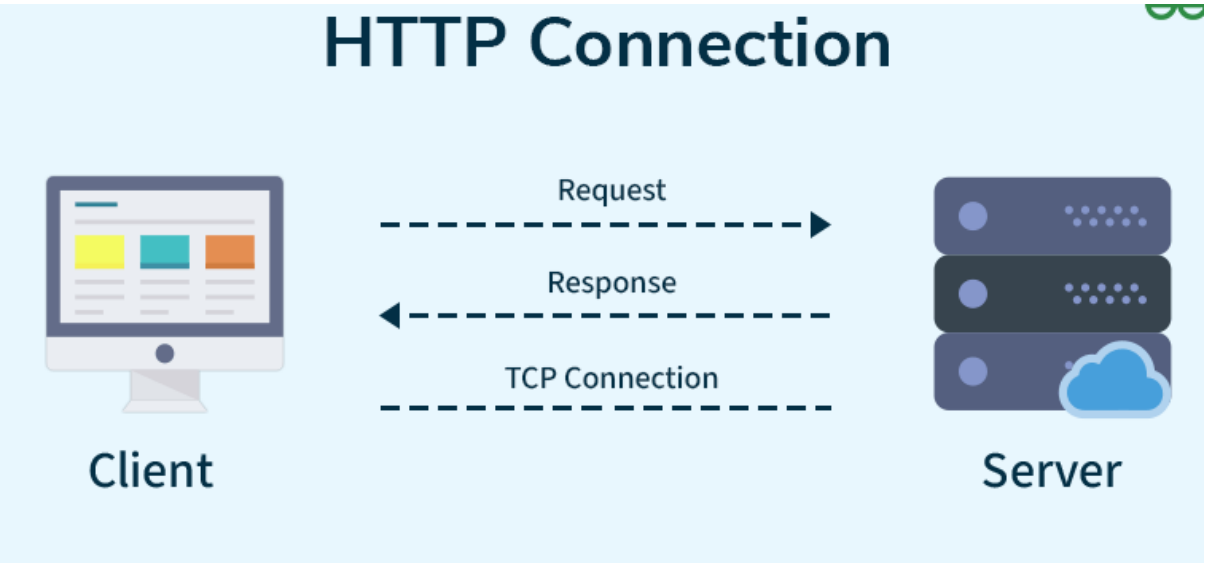
- **HTTP** stands for **Hypertext Transfer Protocol**, and it's the system that allows communication between web browsers (like Google Chrome or Firefox) and websites. When you visit a website, your browser uses HTTP to send a request to the server hosting that site, and the server sends back the data needed to display the page.
Think of HTTP as a set of rules or a language used by your browser and the web server to talk to each other, ensuring that websites load properly when you type in their URLs.

Features of HTTP
The basic features that make HTTP a simple but powerful protocol are mentioned below:

1. **Stateless:** Each request is treated as a new request i.e both client and server forget about each other after completion of a request unless cookies, tokens or sessions are used.
2. **Connectionless:** The connection between client and server is closed after the completion of each request making it a connectionless protocol.
3. **Media Independent**: Any type of data can be transferred over web using HTTP protocol, as long as both the client and server specify the format using MIME types.
4. **HTTP Methods:** HTTP defines various methods for different actions such as GET, POST, PUT and more.
5. **Caching Support**: HTTP provides support for caching which improves the performance by storing the copies of responses and reusing them later.

How HTTP Works: Step-by-Step Process
Here's how HTTP works when you visit a website:
1. **Open Web Browser**: First, you open your web browser and type a website URL (e.g., www.example.com).
1. **DNS Lookup**: Your browser asks a Domain Name System (DNS) server to find out the IP address associated with that URL. Think of this as looking up the phone number of the website.
1. **Send HTTP Request**: Once the browser has the website's IP address, it sends an HTTP **request** to the server. The request asks the server for the resources needed to display the page (like text, images, and videos).
1. **Server Response**: The server processes your request and sends back an HTTP **response**. This response contains the requested resources (like HTML, CSS, JavaScript) needed to load the page.
1. **Rendering the Web Page**: Your browser receives the data from the server and displays the webpage on your screen.

After the page is loaded, the connection between the browser and server is closed. If you request a new page, a new connection will be made.



HTTP messages

HTTP messages are the fundamental units of communication in the Hypertext Transfer Protocol (HTTP), facilitating the exchange of data between a client (such as a web browser) and a server. There are two primary types of HTTP messages: requests and responses.

1. HTTP Request Messages:

A client sends an HTTP request message to a server to request a specific action or resource. It typically consists of:

- **Start Line (Request Line):**
  - **Method:** The HTTP method (e.g., GET, POST, PUT, DELETE) indicating the desired action.
  - **Request-URI:** The Uniform Resource Identifier (URI) of the resource being requested.
  - **HTTP-Version:** The version of HTTP being used (e.g., HTTP/1.1).
- **Headers:** Key-value pairs providing additional information about the request, such as:
  - Host: The domain name of the server.
  - User-Agent: Information about the client software.
  - Accept: Preferred media types for the response.
- **Optional Body:** Data sent with the request, typically used with methods like POST or PUT.

2. <u>HTTP Response Messages</u>:

A server sends an HTTP response message back to the client in reply to a request. It typically consists of:

- **Start Line (Status Line):**
- **HTTP-Version:** The version of HTTP being used.
- **Status-Code:** A three-digit integer indicating the outcome of the request (e.g., 200 for OK, 404 for Not Found).
- **Reason-Phrase:** A human-readable text explanation of the status code.
- **Headers:**

  Key-value pairs providing information about the response, such as:

- Content-Type: The media type of the response body.
- Content-Length: The size of the response body.
- Server: Information about the server software.
- **Optional Body:**

  The actual data or resource requested by the client (e.g., an HTML page, an image).

  Example of an HTTP Request (simplified):

  Code
```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
```

  Example of an HTTP Response (simplified):

  Code
```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1234

<!DOCTYPE html>
<html>
<head>
   <title>Example Page</title>
</head>
<body>
   <h1>Welcome!</h1>
</body>
</html>
```